INTRODUCTION

Research is a very general term for an activity that involves finding out, in a more or less systematic way, things you did not know.

Research is a systematic and logical study of an issue or problem or phenomenon through scientific method.

Research methods are the techniques you use to do research. They represent the tools of the trade, and provide you with ways to collect, sort and analyze information so that you can come to some conclusions.

CHARACTERISTICS OF RESEARCH.

The major characteristics of any research are; Objectivity, precision, design and

verifiability.

Objectivity.

In an ideal situation, a research is beyond the subjective bias of the researcher. As a researcher, you have to make deliberate efforts to eliminate all personal preferences. You should resist the temptation to seek only such data which support your hypotheses or your line of taught. In a scientific research, emphasis is on testing the hypothesis not to prove it. You have to willingly suspend your personal judgment in order to allow the data and logic to lead independently to a sound conclusion. If you want to achieve objectivity in your investigation, you have to use standardized research instruments, choose appropriate research design and analytical tools and also ensure the dependability of data.

Precision

When you conduct a good research and write the report but your readers do not understand what you have done, you may have succeeded in wasting your time and efforts. Every research should use a technical language in order to convey the exact meaning to the readers. Such research languages include validity, reliability, random sampling variables etc. The most precise expression in quantitative research is the mathematical equation or statistical finding which explains or represents the truth. But in qualitative research, precision is achieved through words rather than numbers. So you have to use a very precise language to describe your study accurately. This is done so that the study can be replicated or the results correctly used.

Design

If you want to have a good research, you must have a very good and systematic design. This implies that every scientific inquiry will generally undergo such steps as: -

- Definition of the problem
- Statement of the hypothesis
- Collection and analysis of data
- Testing and confirmation or rejection of hypothesis.
- Reporting of the results.

Any research, which has no orderly design, cannot be replicated for verification.

Verifiability

When you conduct a research, you write your report. This presents the research design and the findings to the professional community. From this point other researchers and scholars will study the report, analyze it in order to confirm or reject the outcomes. This tells you that research is a social enterprise. Its information is open for public scrutiny. Verification is related to objectivity and precision. It is only through further investigation or replication of the study can the results of any study be confirmed, revised or rejected. It is also through this process that a body of new knowledge is developed and new questions identified. Verifiability can be achieved through two different approaches.

- Analyzing the same data on the same sample through alternative analytical tools or statistical methods.
- Replicating the study on a different sample

Research Process

Research process consists of series of actions or steps necessary to effectively carry our research and the desired sequencing of these steps.

Formulating the research problem: The formulation of a general topic into a specific research problem thus constitutes the first step in a scientific enquiry. Essentially two steps are involved in formulating the research problem, viz., understanding the problem thoroughly, and rephrasing the same into meaningful terms from an analytical point of view.

Extensive Literature survey: The abstracting and indexing journals and published or unpublished bibliographies

Formulating the working hypotheses: After extensive literature survey, researcher should state in clear terms the working hypothesis. It is tentative assumption made in order to draw out and test its logical or empirical consequences.

Preparing the research design: Research design includes the means of obtaining the information, explanation of the way in which selected means of obtaining information will be organized and the reasoning leading to the selection.

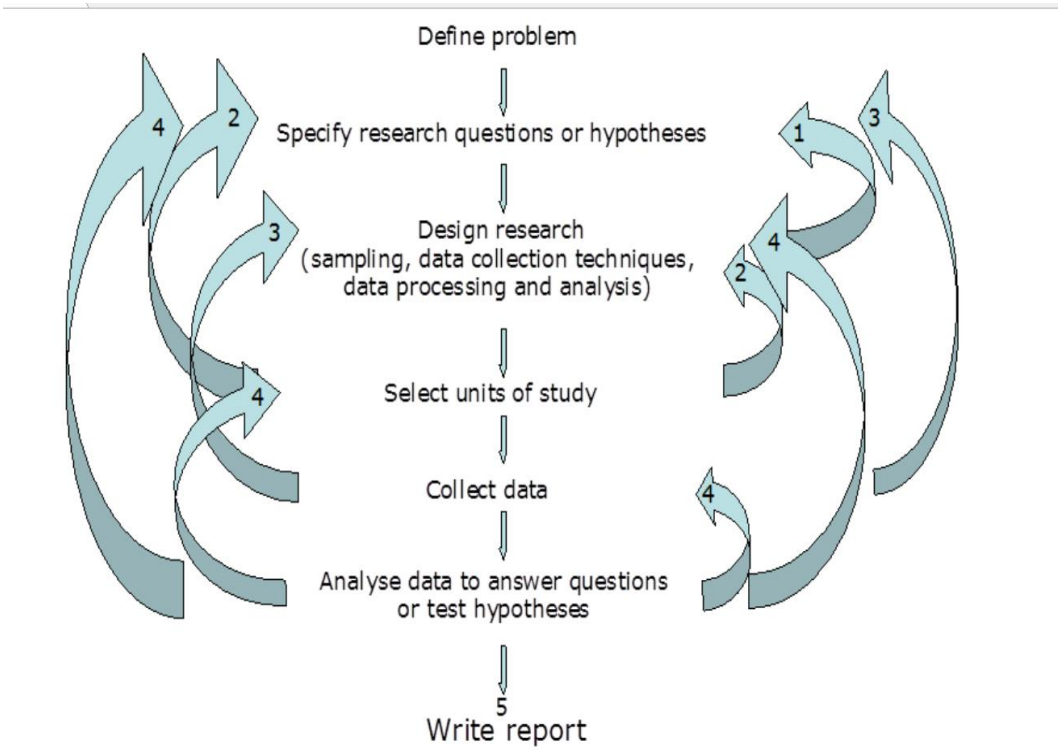
Determining Sample design: A sample design is a definite plan determined before nay data are actually collected for obtaining a sample from a given population.

Collecting the data: Data are two types – Primary data and Secondary data. Primary data can be collected by observation, through personal interview, through telephone interviews, by mailing of questionnaires, through schedules.

Analysis of data: The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences.

Interpret and report: Finally, the researcher has to prepare the report of what has been done by him.

Figure 1: The Research Process



(1) Through the process of designing your research, consideration of both practical and conceptual issues may force you to reconsider your original research question.
(2) Difficulties with access to research sites or participants may cause you to reconsider your questions or your methods.
(3) Issues arising during data collection may suggest that additional data are required or reveal problems with the original research question.
(4) Problems or new questions arising from analysis of data collected so far may result in a need to collect more data, sample elsewhere or employ a different technique.
(5) Finally you progress to the report writing stage.

Regardless of the route taken subsequently, research should start with the problem and the research questions. If the intention of research is to answer your questions, it follows that choice of method should develop from your question: choose the method that can best provide the information you need to answer your research question given the resources available to you. This is one reason why it is very important to be clear as to what you are asking. Planning your research involves the consideration of four overlapping themes.

- The conceptual approach – the philosophical underpinnings of research
- Research design – how data collection is organized
- Data collection techniques – how data are collected
- Sampling – from whom data are collected

PURPOSE OF RESEARCH

Training in Research.

By going through this course, you are subjecting yourself to be trained on how to conduct researches. Executing a research project in a scientific manner needs training in research methods. This course on research methodology provides the theoretical background for your research project. The course will facilitate the right choice of research designs, tools and techniques.

Problem solving.

This involves diagnosing and solving the problems in the system. Problem solving means a realistic understanding of a situation on the basis of data and statistics. Eg: You can take studies on the various aspects of the electronic business and utilization and show the differential out these using electronics and these who do not. What we are saying is that a series of studies can go into the diagnosis in order to develop better understanding of the practices in a system. You may also set out studies to actually get into experimenting with an innovative solution. You can device various methods of receiving customers and making their comfortable.

Search for Truth.

This is the biggest challenge in research. A research in search of truth is most of the times classified as pure research. It is the ability to generalize and create knowledge. Such generalizations are derived from occurrences in repeated instances. Generalizations are drawn primarily on two bases. These are repeated observations in various possible situations and applications of statistical designs where the variables are statistically controlled.